

3.5.7 DG: Binary operations

	Content
DG1	Understand and use binary operations including use of modular arithmetic and matrix multiplication.

	Content
DG2	Understand, use and prove the commutativity of a binary operation.

	Content
DG3	Understand, use and prove the associativity of a binary operation.

	Content
DG4	Construct a Cayley table for a given set under a given binary operation.

	Content
DG5	Understand and prove the existence of an identity element for a given set under a given binary operation.

	Content
DG6	Find the inverse of an element belonging to a given set under a given binary operation.

	Content
DG7	Understand and use the language of groups including: order, period, subgroup, proper, trivial, non-trivial.

	Content
DG8	Understand and use the group axioms: closure, identity, inverses and associativity, including use of Cayley tables.

	Content
DG9	Recognise and use finite and infinite groups and their subgroups, including: groups of symmetries of regular polygons, cyclic groups and abelian groups.

	Content
DG10	Understand and use Lagrange's theorem.

	Content
DG12	Recognise and find isomorphism between groups of finite order.